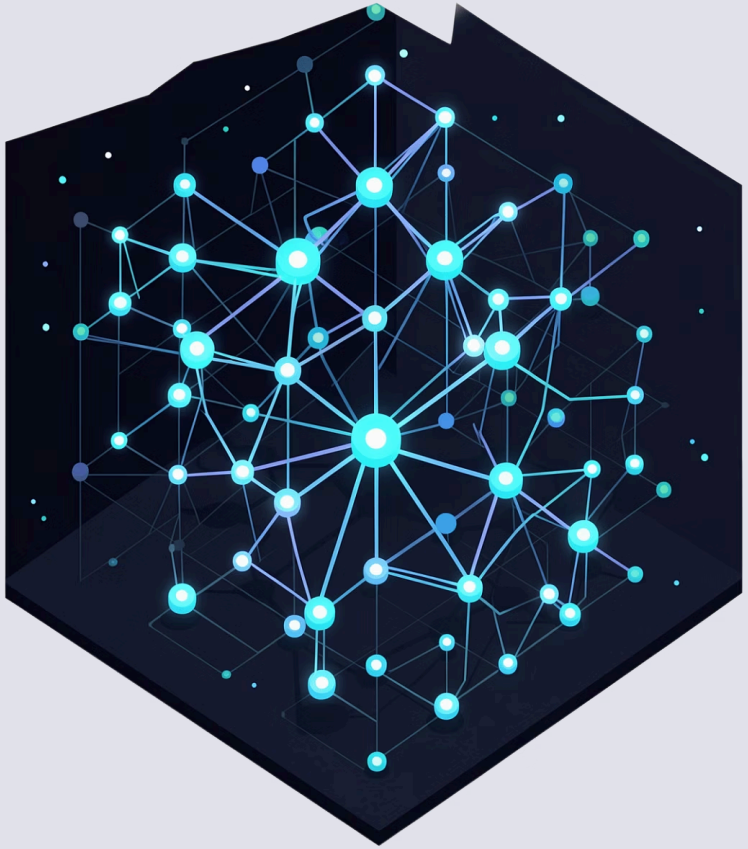
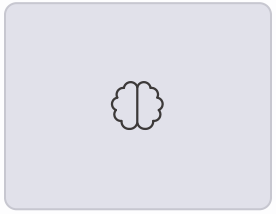


Foundation Models & Generative AI

GPT, BERT — Architectures, Training & Industry Applications

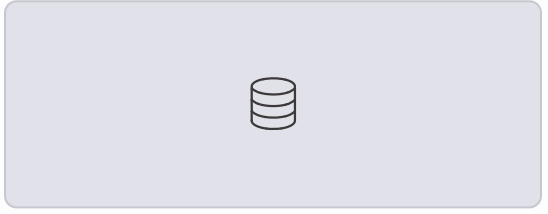
Dr. Anilkumar V Brahmane | Computer Engineering Department, Sanivani College of Engineering

What is Artificial Intelligence? — A Quick Recap



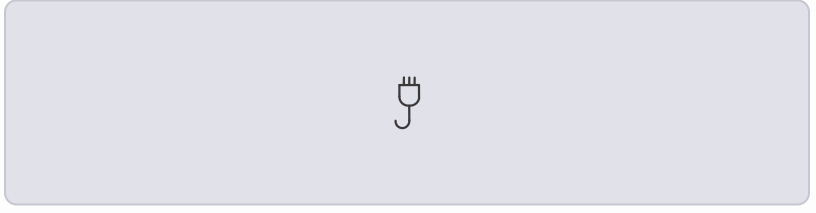
AI

Machines that simulate human intelligence — reasoning, learning, problem-solving



Machine Learning

AI that learns from data without being explicitly programmed



Deep Learning

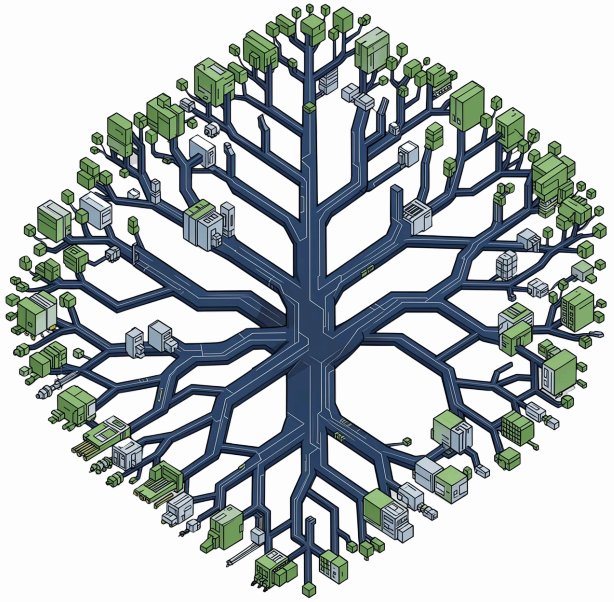
Multi-layered neural networks that learn complex patterns



Generative AI

AI that can *create* new content — text, images, code, audio

What Are Foundation Models?



The Core Idea

Definition: Massive AI models pre-trained on huge datasets, adaptable to many tasks

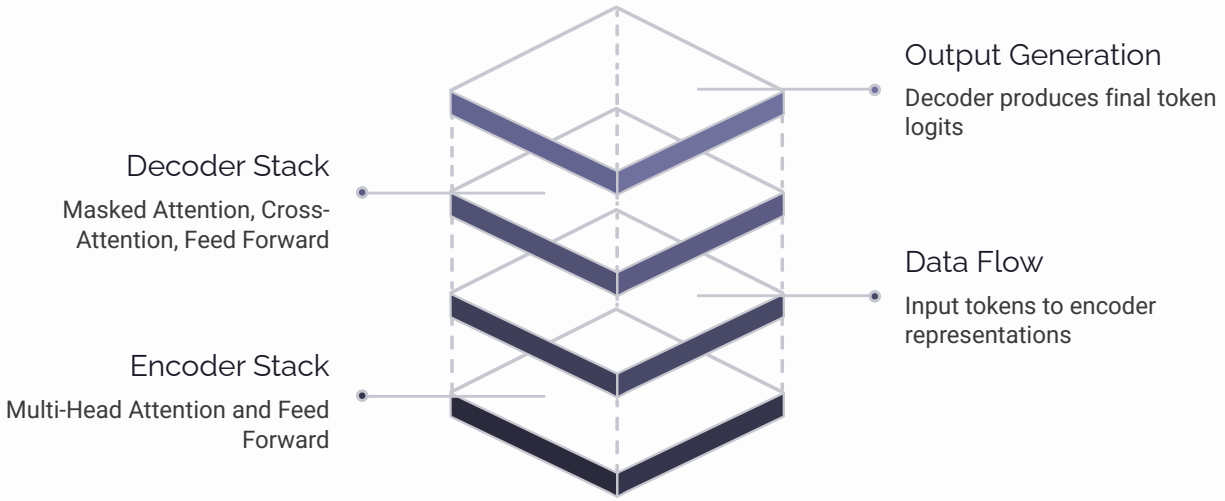
Scale: Billions of parameters trained on terabytes of text, images, and code

Key idea: Train once → Fine-tune for many downstream tasks (translation, Q&A, coding)

Examples: GPT-4, BERT, LLaMA, PaLM, Gemini, DALL-E

One Foundation Model → Many Applications: Translation, Q&A, Summarization, Code Generation, Image Generation

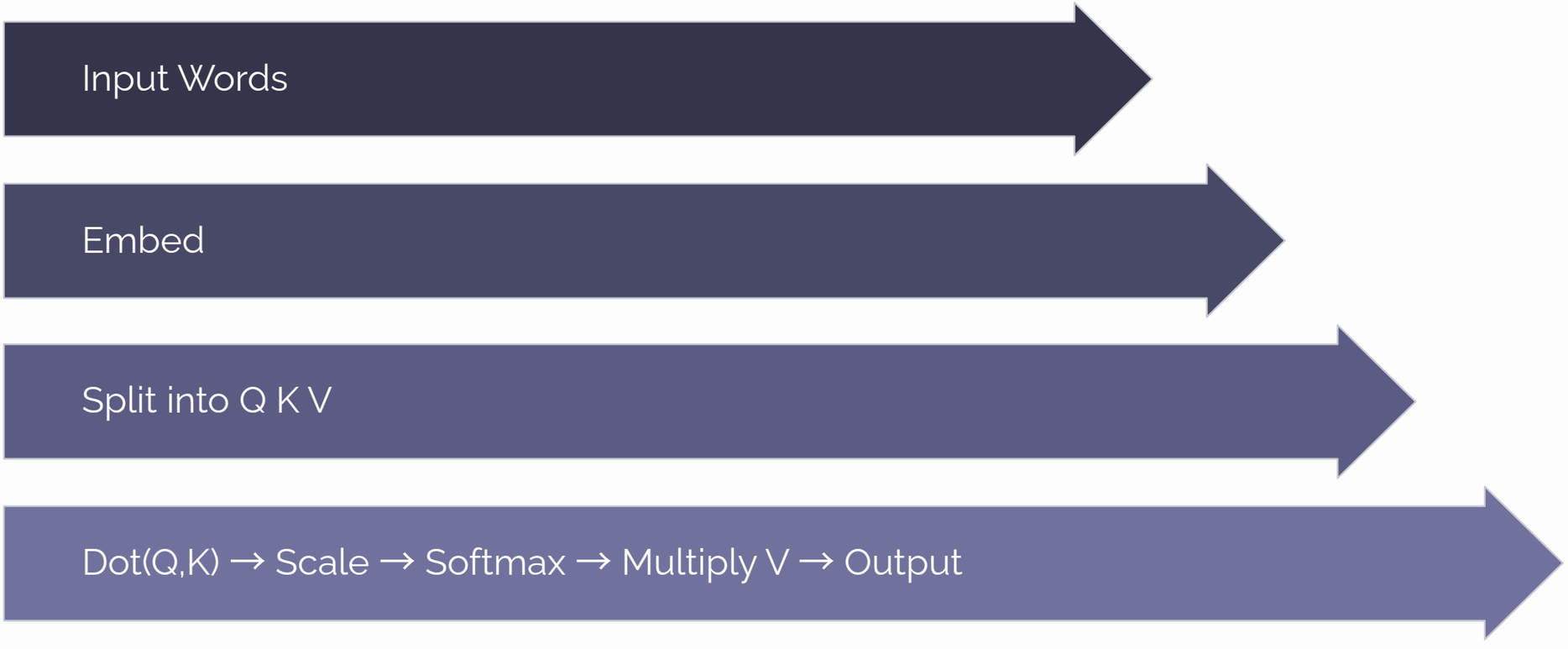
The Transformer — Heart of All Foundation Models



Key Facts

- Introduced in 2017 paper *"Attention Is All You Need"* by Google
- **Key innovation:** Self-Attention — model focuses on relevant words simultaneously
- Processes all words **in parallel** (unlike RNNs which process one-by-one)
- Two main blocks: **Encoder** (understands input) + **Decoder** (generates output)

Self-Attention — How Transformers "Understand" Language



Simple Intuition

Simple idea: When reading a word, look at ALL other words to understand its meaning

Example: "The **bank** by the river" — attention links "bank" to "river" (not money)

Q, K, V Mechanism

Query, Key, Value (Q, K, V): Three vectors computed for each word to calculate attention scores

Multi-Head Attention: Run attention multiple times in parallel to capture different relationships

GPT — Generative Pre-trained Transformer

Developer

OpenAI (2018 → GPT-2 → GPT-3 → GPT-4)

Architecture

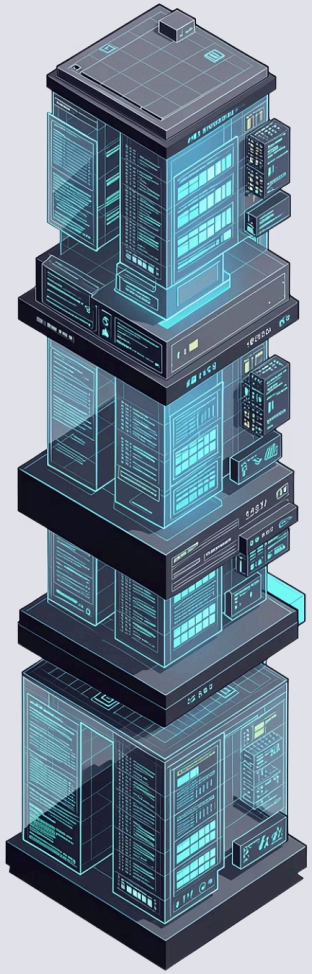
Decoder-only Transformer — reads left to right

Training Objective

Causal Language Modeling — predict the *next word* given previous words

Scale

GPT-3: 175B parameters; GPT-4: multimodal (text + images)



How GPT is Trained — Step by Step



The 5-Stage Pipeline

01

Collect Data

Billions of web pages, books, Wikipedia (Common Crawl dataset)

02

Tokenize

Break text into tokens using BPE (Byte Pair Encoding)

03

Pre-train

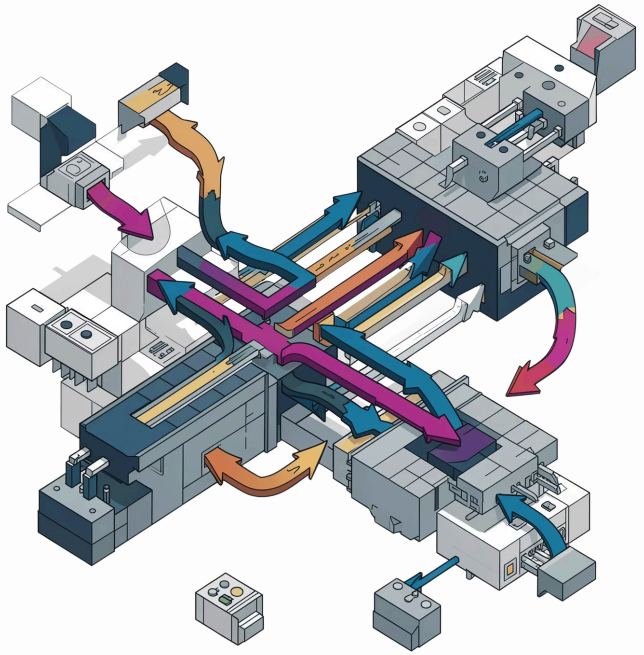
Predict next token — model learns grammar, facts, reasoning

04

Fine-tune + RLHF

Humans rank outputs to align model behavior

BERT — Bidirectional Encoder Representations from Transformers

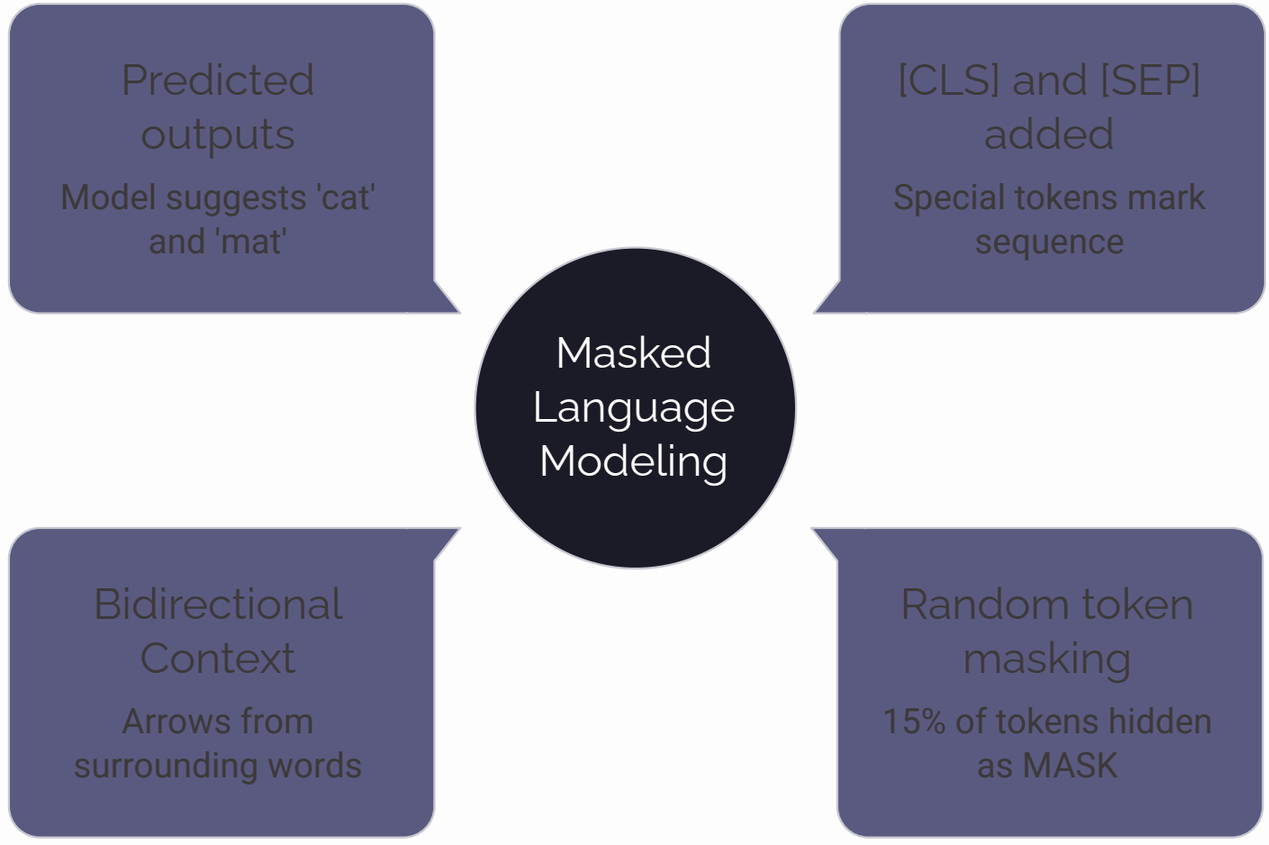


Overview

- **Developed by:** Google (2018) — published at NAACL 2019
- **Architecture:** Encoder-only Transformer — reads in *both directions* simultaneously
- **Training objective:** Masked Language Modeling (MLM) + Next Sentence Prediction (NSP)
- BERT-Base: 110M parameters; BERT-Large: 340M parameters

> Key insight: BERT predicts a randomly hidden token using bidirectional context — both left and right sides of the sentence at once.

How BERT is Trained — Step by Step



1 Input Preparation

Add special tokens [CLS] (start) and [SEP] (separator)

2 Masking

Randomly mask 15% of tokens — model must predict the hidden words

3 Bidirectional Reading

Model looks at words on BOTH left and right sides simultaneously

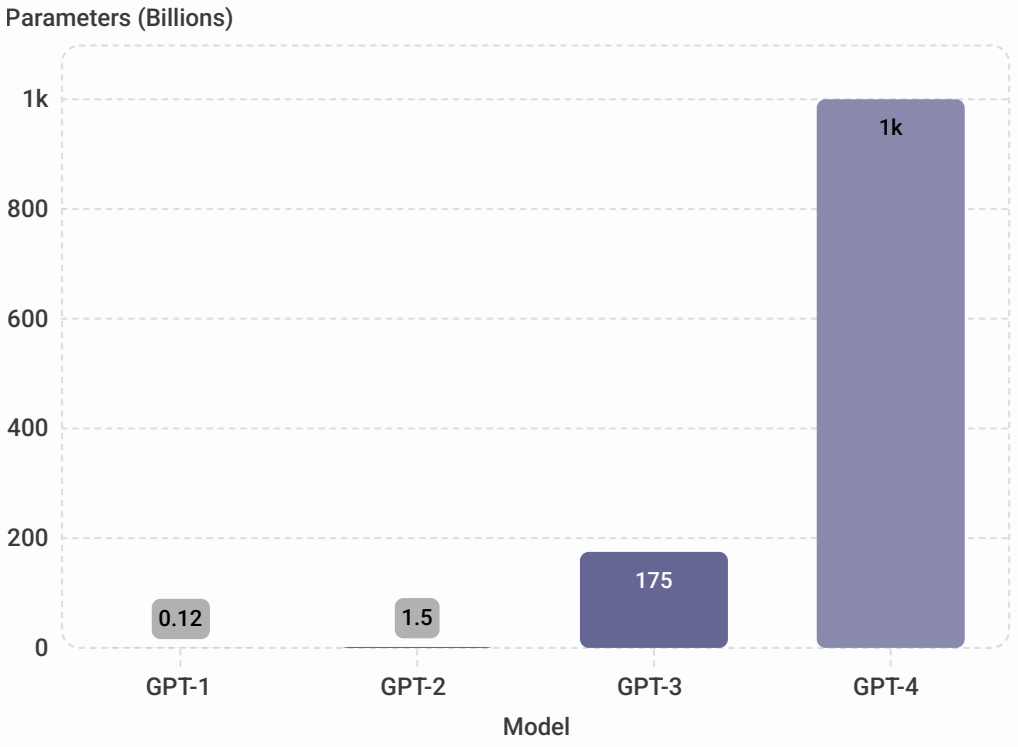
4 NSP + Fine-tune

Predict if Sentence B follows A; then add task layer for Q&A, NER, Classification

GPT vs BERT — Side-by-Side Comparison

Feature	GPT	BERT
Developer	OpenAI	Google
Architecture	Decoder-only	Encoder-only
Reading Direction	Left → Right (Causal)	Bidirectional (Both ways)
Training Task	Next Token Prediction	Masked Language Model + NSP
Best For	Text Generation, Chatbots	Understanding, Classification, Q&A
Example Use	ChatGPT, Copilot	Google Search, NER, Sentiment

Scaling Laws — Why Bigger Models Are Smarter



The Scaling Principle

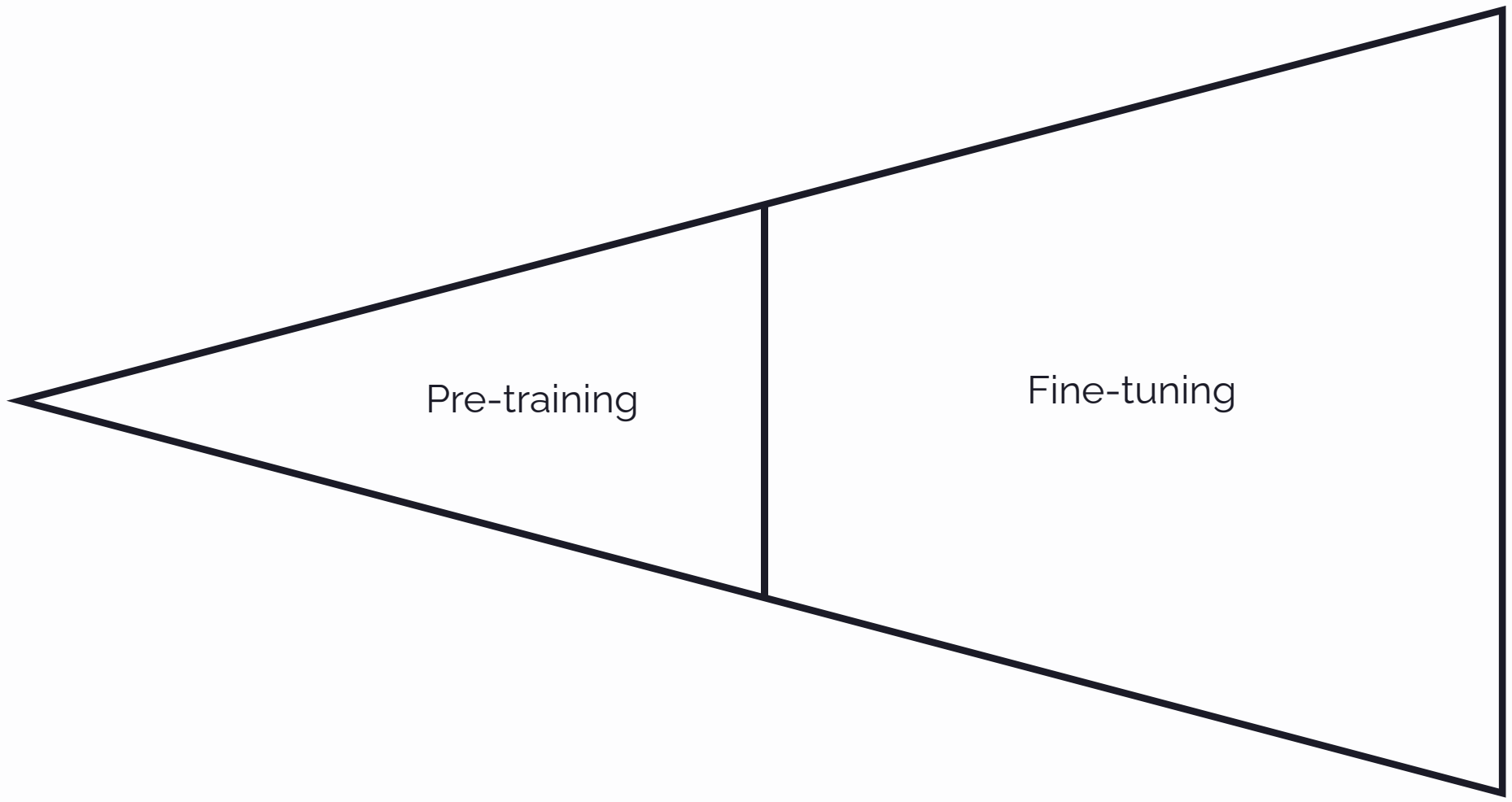
Scaling Law: More parameters + more data + more compute = better performance

Model size has grown **exponentially** — from 117M (GPT-1) to an estimated ~1 Trillion parameters (GPT-4).

Emergent abilities: At large scale, models suddenly gain new skills — reasoning, coding — not seen in smaller models.

Trade-off: Larger models require massive GPU clusters and significant energy consumption.

Pre-training vs Fine-tuning — The Two-Stage Approach



Pre-training

Learn general knowledge from massive **unlabeled** data (self-supervised learning on web text, books, code)

Fine-tuning

Adapt the pre-trained model to a **specific task** using a small labeled dataset (e.g., medical Q&A, legal docs)

Key Advantage

No need to train from scratch for every new task — saves enormous time, cost, and compute resources

Generative AI — What Can It Create?



Text

Articles, emails, summaries, stories

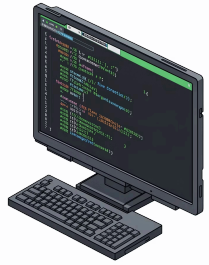
GPT-4, LLaMA



Images

Photorealistic art, diagrams, logos

DALL-E, Stable Diffusion, Midjourney



Code

Write, debug, explain programs

GitHub Copilot, CodeLlama



Audio & Video

Music, voice cloning, video synthesis

Sora, ElevenLabs

Applications in Healthcare

Medical Diagnosis

AI reads X-rays, MRIs, and pathology slides with high accuracy

Drug Discovery

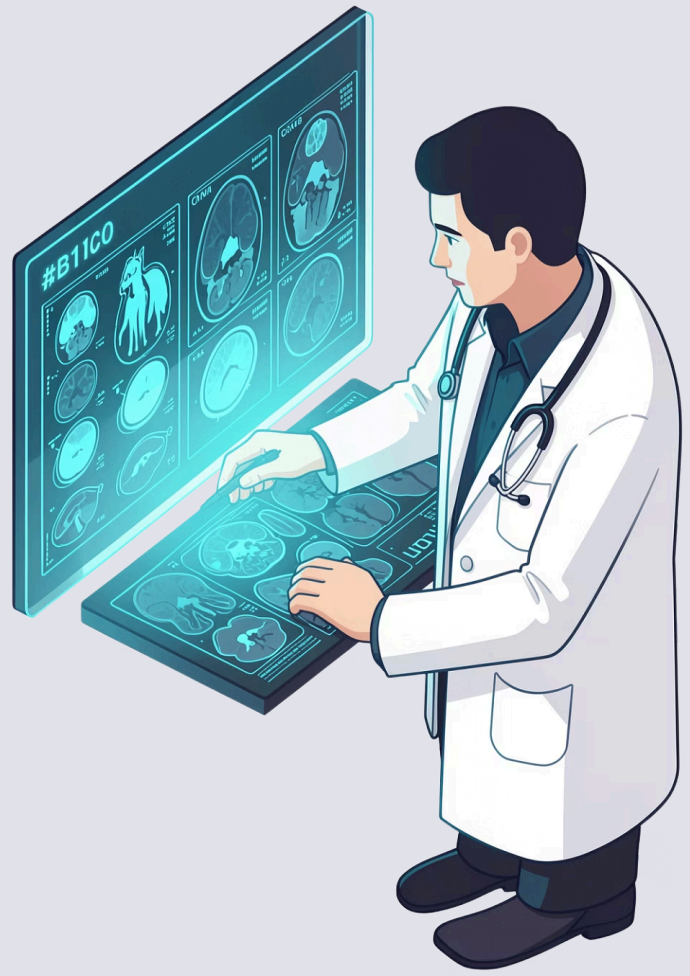
GPT-based models predict protein structures and drug interactions (AlphaFold)

Clinical Notes

Auto-generate patient summaries from doctor-patient conversations

AI Chatbots

24/7 patient support, symptom checking, appointment scheduling



Applications in Finance & Education

Finance

Fraud Detection

LLMs analyze transaction patterns to flag anomalies in real time

Report Generation

Auto-generate earnings summaries, risk assessments, and compliance docs

Education

Personalized Tutoring

AI adapts explanations to each student's level (Khan Academy AI)

Auto-Grading

Evaluate essays, code assignments, and provide instant feedback

Applications in Software Engineering & Retail



Software Engineering

- **Code Generation:** GitHub Copilot writes 40%+ of code for developers using GPT-4
- **Bug Detection:** AI reviews code, identifies vulnerabilities, suggests fixes

Retail

- **Personalization:** Recommend products based on browsing history and preferences
- **Customer Service:** AI chatbots handle returns, queries, and complaints 24/7

Organizations report a 7.8% improvement in productivity and 6.7% improvement in customer engagement from generative AI adoption.

Challenges & Limitations of Foundation Models

👻 Hallucinations

Models confidently generate **false or made-up information** — dangerous in high-stakes domains

⚖️ Bias

Training data contains societal biases — the model inherits and **amplifies** them

💰 High Cost

Training GPT-3 cost **~$4.6 million** in compute; inference at scale is also expensive

🔒 Privacy Risk

Models may memorize and **leak sensitive training data** — a serious compliance concern

❓ Black Box

Hard to understand **why** a model gave a specific answer — poor explainability

Ethical AI & Responsible Use



The 4 Pillars of Responsible AI

Fairness

Audit models for bias before deployment in critical systems

Transparency

Disclose when content is AI-generated (watermarking, labels)

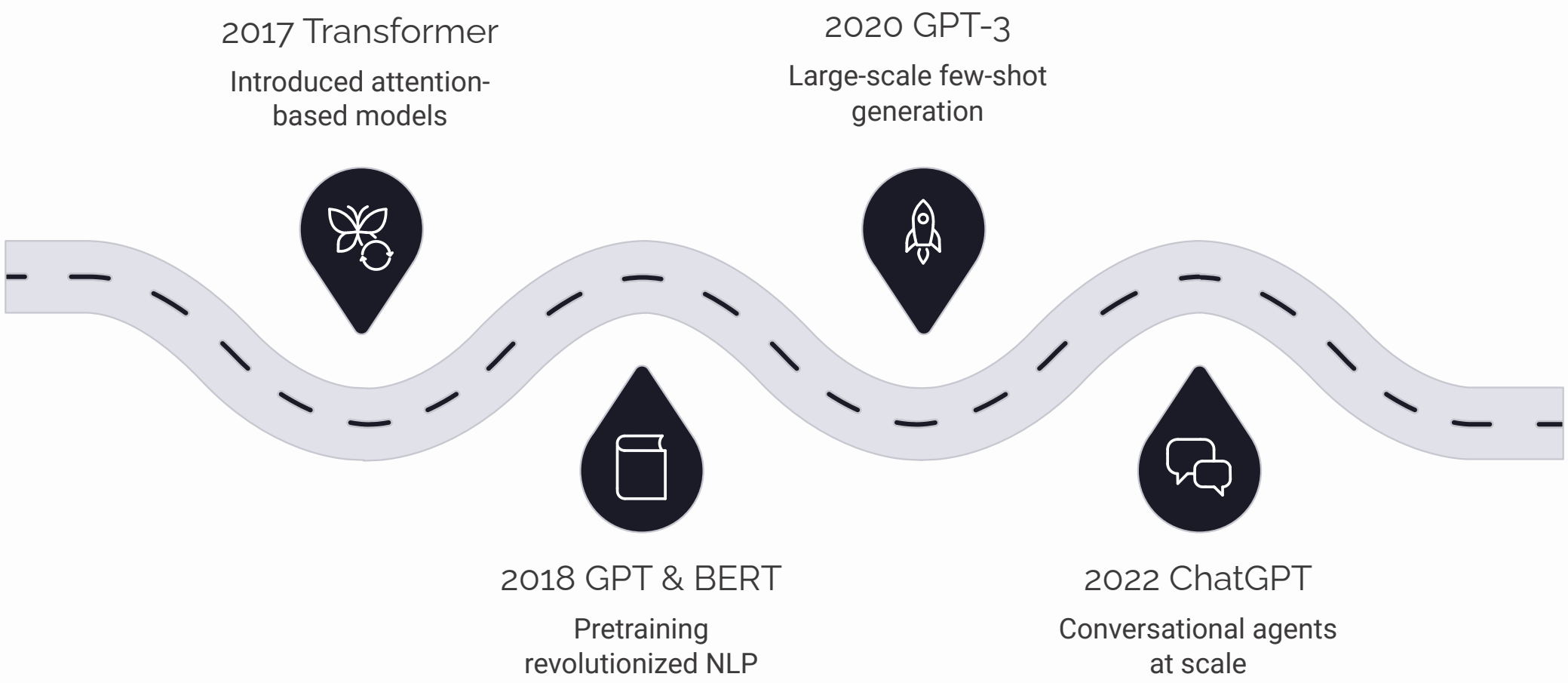
Regulation

EU AI Act (2024) classifies AI by risk level and mandates compliance

Human Oversight

Keep humans in the loop for high-stakes decisions (medical, legal)

Future of Foundation Models



Multimodal AI

Models understanding text + image + audio + video together (GPT-4o, Gemini)

Smaller & Efficient

Distillation techniques create compact models for mobile and edge devices

AI Agents

Autonomously plans and executes multi-step tasks (AutoGPT, LangChain)

Domain-Specific

Med-PaLM (medicine), BloombergGPT (finance), CodeLlama (coding)



Key Takeaways

Foundation Models

Large pre-trained models adaptable to many tasks via fine-tuning — train once, deploy everywhere

GPT = Generation

Decoder-only, generates text left-to-right; best for chatbots, code, and creative tasks

BERT = Understanding

Encoder-only, reads bidirectionally; best for search, classification, and Q&A

Responsible AI is Essential

Fairness, transparency, and human oversight must guide every deployment

Dr. Anilkumar V Brahmane | Computer Engineering Department, Sanivani College of Engineering

Sources

- **Chapter 41: Pretraining — BERT and GPT — Classical Foundations of Artificial Neural Networks**
- **Generative AI in Organizations — Capgemini Research Institute (2024)**